Practical Session 2 – Exploratory Data Analysis and Statistics

Big Data Course – Third-Year Undergraduate (L3)



Learning Objectives

This second practical session aims to help you:

- develop your data manipulation skills using `pandas` and `numpy`;
- carry out a complete exploratory data analysis;
- produce relevant data visualizations;
- critically interpret your results.

This practical session is **not graded**, but submission on Community is **mandatory**. It directly prepares you for the **course project**. Submitting your work allows us to assess your engagement in the learning process and provide individual feedback.

Working Environment

Your work must be completed in a **Jupyter notebook** using Python and the following libraries:

- `pandas`
- `numpy`
- `matplotlib` and/or `seaborn`

You may reuse the online Jupyter environment or a local environment.

Dataset

You will use the **dataset cleaned during Practical Session 1**. If necessary, you may improve or correct your previous cleaning work.

Tasks

1. Loading the Data

- Load the cleaned dataset.
- Check the overall consistency of the data.

2. Descriptive Statistics

Calculate and comment on:

- basic statistics (mean, median, minimum, maximum);
- measures of dispersion (variance, standard deviation);
- other statistics relevant to the dataset.

3. Exploratory Data Analysis

Perform an exploratory analysis of the dataset to identify patterns or findings of interest. In particular, your analysis should include:

- selecting one or more quantitative variables and studying how they change;
- identifying overall trends or significant variations;
- comparing data according to at least one relevant criterion (for example: categories, groups, time periods, or geographic areas).

Each step of the analysis must include a short comment explaining your choices and the results observed.

4. Visualizations

Produce appropriate visualizations, such as:

- histograms;
- line charts showing changes over time;
- bar charts;
- scatter plots.

Each figure must:

- have a title;
- have properly labeled axes;
- be accompanied by a short interpretation.

5. Research Question

Formulate a **simple question** that can be answered using the data, for example:

- How does a particular variable change over time?
- Is there a significant difference between two groups?

For example, you could investigate the following questions:

- Which New Year's resolution category is the most (or least) common?
- Which resolution category received the most retweets?
- At what time of day are resolutions most frequently tweeted?
- Which U.S. region or state posts the most resolutions?

Provide a reasoned answer to one main question. You may include additional analyses if they are relevant.

Expected Deliverables

You must submit:

- a clear, well-structured Jupyter notebook;
- readable charts accompanied by comments;
- a short conclusion summarizing your main findings.

Important Notes

- This practical session is not graded, but submission is mandatory.
- Clear code and comments are essential.
- Results must be interpreted and discussed, not merely presented.
- This work provides a direct foundation for the **course project**.